

File Explorer

- docker practice
 - terraform practice
 - .terraform
 - .terraform.lock.hcl
 - ec2.tf
 - iam.tf
 - provider.tf
 - terraform.tfstate
 - terraform.tfstate.backup
 - vpc.tf

Outline

Timeline

```
terraform practice > ec2.tf > resource "aws_instance" "testec2" > associate_public_ip_address
22     }
23
24     resource "aws_vpc_security_group_ingress_rule" "https1" {
25         security_group_id = aws_security_group.ec2-sg.id
26         cidr_ipv4         = "0.0.0.0/0"
27         from_port         = 443
28         ip_protocol       = "tcp"
29         to_port           = 443
30     }
31     resource "aws_vpc_security_group_ingress_rule" "http1" {
32         security_group_id = aws_security_group.ec2-sg.id
33         cidr_ipv4         = "0.0.0.0/0"
34         from_port         = 80
35         ip_protocol       = "tcp"
36         to_port           = 80
37     }
38     resource "aws_vpc_security_group_ingress_rule" "ssh1" {
39         security_group_id = aws_security_group.ec2-sg.id
40         cidr_ipv4         = "0.0.0.0/0"
41         from_port         = 22
42         ip_protocol       = "tcp"
```

Problems Output Debug Console Terminal Ports

powershell - terraform practice + -

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply
aws_vpc_security_group_ingress_rule.ssh: Refreshing state... [id=sgr-0ad20ee409e37ca0e]
aws_vpc_security_group_ingress_rule.http: Refreshing state... [id=sgr-00bce0519a8f4fd9f]
aws_vpc_security_group_ingress_rule.https1: Refreshing state... [id=sgr-028fa814645bf09a0]
aws_instance.testec2: Refreshing state... [id=i-0e17cc865844fe44a]
aws_route_table_association.demo-association: Refreshing state... [id=rtbassoc-0855ab976454edd65]
```

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice>

```
terraform practice > ec2.tf > resource "aws_instance" "testec2" > ami
```

```

1 resource "aws_instance" "testec2" {
2     ami           = var.ami-reference
3     instance_type = "t3.micro"
4     key_name      = "key1"
5     subnet_id     = aws_subnet.subnet1.id
6     vpc_security_group_ids = [aws_security_group.ec2-sg.id]
7     associate_public_ip_address = true
8
9     tags = {
10         Name = "demo1ec2"
11         name = "demoinstance"
12         team = "sjce-devops"
13     }
14 }
15 resource "aws_security_group" "ec2-sg" {
16     name           = "allow_tls1"
17     description    = "Allow TLS inbound traffic and all outbound traffic"
18     vpc_id         = aws_vpc.vpc.id
19     tags = {
20         Name = "Ec2-sg"
21     }

```

Problems Output Debug Console **Terminal** Ports

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
```

Enter a value: yes

```
aws_instance.testec2: Creating...
aws_instance.testec2: Still creating... [00m10s elapsed]
aws_instance.testec2: Still creating... [00m20s elapsed]
aws_instance.testec2: Creation complete after 20s [id=i-01dfdc3352683cef7f]
```

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice>
```

> Timeline

EC2

Dashboard
AWS Global View
Events

Instances

Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans

Successfully initiated termination (deletion) of i-0e17cc865844fe44a

Instances (2) Info

Last updated
less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Filter Instance by attribute or tag (case-sensitive)

< 1 >



☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	demo1ec2	i-0e17cc865844fe44a	Terminated	t3.micro	–	us-east-1c	–
☐	demo1ec2	i-01dfd3352683cef7f	Running	t3.micro	Initializing	us-east-1c	–

VPC dashboard <

Filter by VPC ▼

[AWS Global View](#) ↗

▼ Virtual private cloud

[Your VPCs](#)

[Subnets](#)

[VPCs](#)

VPC encryption controls

Your VPCs (2) [Info](#)

Last updated
less than a minute ago



Actions ▼

Create VPC

Find VPCs by attribute or tag



1



☐	Name ▼	VPC ID ▼	State ▼	Encryption c... ▼	Encryption control ... ▼	Block Public... ▼	IP
☐	-	vpc-02059d2e76a4c51d2	✓ Available	-	-	⊖ Off	17
☐	demo-vpc	vpc-0dba337ae20d4a630	✓ Available	-	-	⊖ Off	10

▼ docker practice

▼ terraform practice

> .terraform

≡ .terraform.lock.hcl

ec2.tf

iam.tf

provider.tf

{ } terraform.tfstate

≡ terraform.tfstate.backup

terraform.tfvars

variable.tf

vpc.tf

provider.tf

ec2.tf

terraform.tfvars X

vpc.tf

variable.tf

terraform practice > terraform.tfvars > abc instance_type

1 ami-reference = "ami-0b6d9d3d33ba97d99"

2 instance_type = "t3.micro"

Problems

Output

Debug Console

Terminal

Ports

powershell

PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply

- volume_type = "gp3" -> null

(1 unchanged attribute hidden)

}

~ secondary_network_interface (known after apply)

}

Plan: 1 to add, 0 to change, 1 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.testec2: Destroying... [id=i-01dfd3352683cef7f]

aws_instance.testec2: Still destroying... [id=i-01dfd3352683cef7f, 00m10s elapsed]

aws_instance.testec2: Destruction complete after 11s

aws_instance.testec2: Creating...

aws_instance.testec2: Still creating... [00m10s elapsed]

aws_instance.testec2: Creation complete after 17s [id=i-048cb2ce8628a2eab]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice>

> Outline

> Timeline

docker practice

terraform practice

> .terraform

≡ .terraform.lock.hcl

ec2.tf

iam.tf

outputs.tf

provider.tf

{ } terraform.tfstate

≡ terraform.tfstate.backup

terraform.tfvars

variable.tf

vpc.tf

terraform practice > outputs.tf > output "ec2_public_ip"

```
1 output "ec2_public_ip" {  
2   value = aws_instance.testec2.public_ip  
3 }
```

Problems Output Debug Console Terminal Ports

powershell - terraform practice + -

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aws_vpc_security_group_ingress_rule.https1: Refreshing state... [id=sgr-028fa814645bf09a0]  
aws_vpc_security_group_ingress_rule.ssh1: Refreshing state... [id=sgr-0228b82dbea6d0182]  
aws_vpc_security_group_ingress_rule.http1: Refreshing state... [id=sgr-057dffeeb5815cdd9]  
aws_instance.testec2: Refreshing state... [id=i-048cb2ce8628a2eab]  
aws_route_table_association.demo-association: Refreshing state... [id=rtbassoc-0855ab976454edd65]
```

Changes to Outputs:

+ ec2_public_ip = "54.234.108.168"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

ec2_public_ip = "54.234.108.168"

○ PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> |

Outline

Timeline

EC2

> Instances

Dashboard

AWS Global View

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Instances

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Instance Types

Launch Templates

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Instances (3)

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Last updated less than a minute ago

Refresh

Connect

Instance state

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Choose filter set

Filter Instance by attribute or tag (case-sensitive)

Name

demo1ec2

i-0e17cc865844fe44a

Terminated

t3.micro

–

us-east-1c

–

demo1ec2

i-01dfd3352683cef7f

Terminated

t3.micro

–

us-east-1c

–

demo1ec2

i-048cb2ce8628a2eab

Running

t3.micro

3/3 checks passed

us-east-1c

–

